dimensions. Newton didn't just say that the force of attraction between two objects gets weaker as the distance between them gets larger. He proposed a formula, the *inverse square law,* which describes precisely how the gravitational attraction will diminish as two objects are separated. According to this formula, if you double the distance between the objects, their gravitational attraction will fall by a factor of 4 (2^2); if you triple the distance, it will fall by a factor of 9 (3^2); if you quadruple the distance, it will fall by a factor of 16 (4^2); and more generally, the gravitational force drops in proportion to the square of the separation. As has become abundantly evident over the last few hundred years, this formula works.

But *why* does the force depend on the square of the distance? Why doesn't the force drop like the cube of the separation (so that if you double the distance, the force diminishes by a factor of 8) or the fourth power (so that if you double the distance, the force diminishes by a factor of 16), or perhaps, even more simply, why doesn't the gravitational force between two objects drop in direct proportion to the separation (so that if you double the distance, the force diminishes by a factor of 2)? The answer is tied directly to the number of dimensions of space.

One way to see this is to think about how the number of gravitons emitted and absorbed by the two objects depends on their separation, or by thinking about how the curvature of spacetime that each object experiences diminishes as the distance between them increases. But let's take a simpler, more old-fashioned approach, which gets us quickly and intuitively to the correct answer. Let's draw a figure (Figure 13.4a) that schematically illustrates the gravitational field produced by a massive object—let's say the sun—much as Figure 3.1 schematically illustrates the magnetic field produced by a bar magnet. Whereas magnetic field lines sweep around from the magnet's north pole to its south pole, notice that gravitational field lines emanate radially outward in all directions and just keep on going. The strength of the gravitational pull another object— imagine it's an orbiting satellite—would feel at a given distance is proportional to the density of field lines at that location. The more field lines penetrate the satellite, as in Figure 13.4b, the greater the gravitational pull to which it is subject.